

Research software development funding calls

Chris Johnson (EPCC), Neil Chue Hong (EPCC and SSI) and Lorna Smith
(EPCC)

17th June 2026

<https://www.archer2.ac.uk/ecse/>



Introduction



- Development and maintenance of software is clearly key to the efficient use of compute resources
 - Especially so with large and relatively complex HPC systems
- Changes in underlying technology require software updates
 - E.g. updates to hardware, libraries, compilers, memory models, programming models, etc.
 - The bigger the underlying changes, the more effort and expertise is required – even just to keep current simulations running...
 - ...further effort required to introduce new algorithms, new science, etc.
- **Software development is at least as important as new hardware!**

Software development funding provision



- Software development funding may be provided and administered in different ways
- Calls may be administered via Research Councils, or via other organisations
- Focus here is on two recent/current programmes
 - eCSE (embedded Computational Science and Engineering)
 - RSMF (Research Software Maintenance Fund)
- We will also make mention of other schemes

Aims of session

- ~30 mins: short talks of the eCSE and RSMF programmes
- ~45 mins of interactive session – we will give you some questions to consider (not restricted to these!)
- Overall aim is to collect feedback on the points raised in the questions
- Additionally we will gather list of priorities based on discussion
- Final aim (depending on time) – Q&A session at end where speakers can comment on possible implications
 - Based on knowledge of recent, current landscape, etc.
- Funding comes from RCs and exactly what to fund and when is at the UKRI level – however, we do have some say in how calls are run

Research software development funding calls: eCSE Calls

Chris Johnson, EPCC

17th June 2026

<https://www.archer2.ac.uk/ecse/>



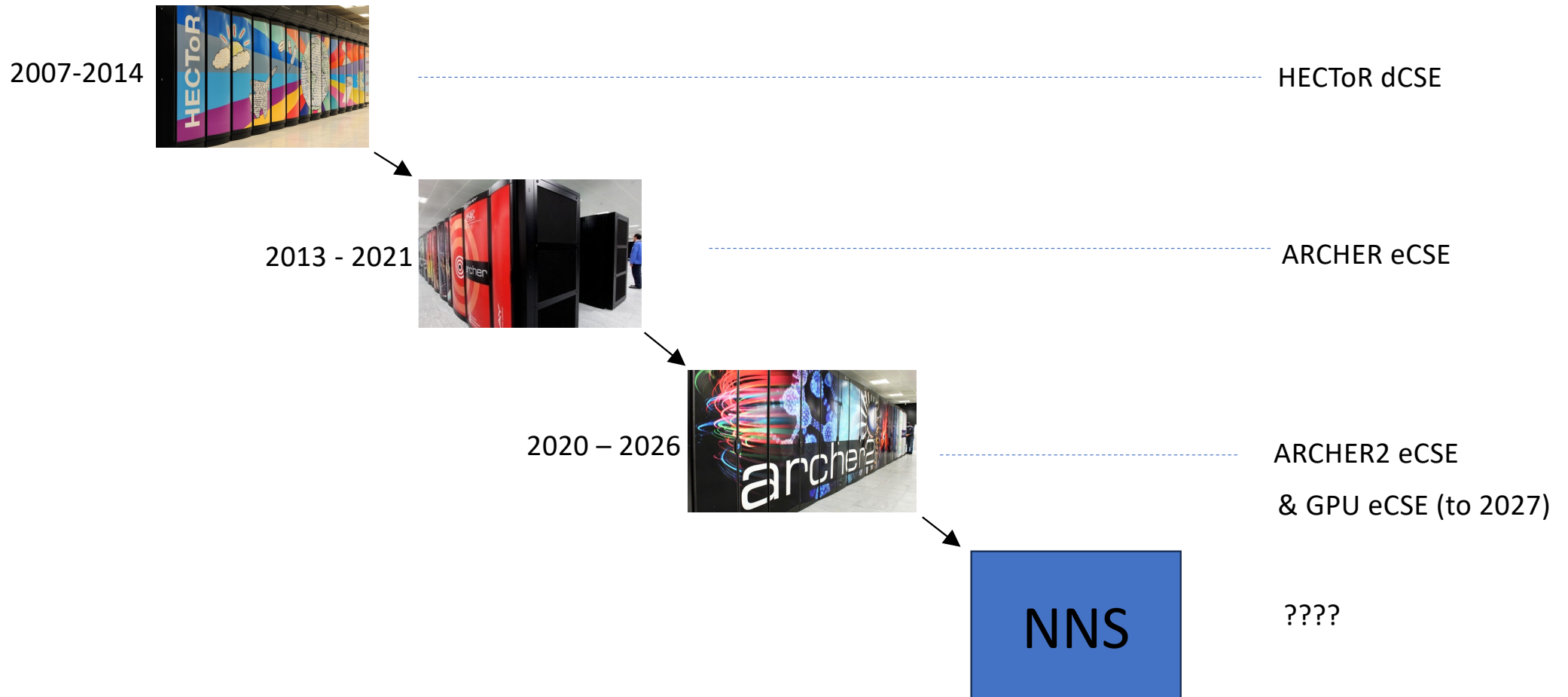
What is the eCSE programme?



- A software development programme that funds RSEs
- Recent calls were administered by EPCC through ARCHER2 contract
- Funding “embedded” across UK HPC community via open calls
 - Funding has been provided to RSE across ~40 Institutions
 - ...more when institutions when PIs and Co-Is included



Evolution of eCSE programme



eCSE remit

- Early calls funded software to run on ARCHER/ARCHER2
 - Remit: EPSRC (& NERC for ARCHER and some ARCHER2 calls)
 - For codes running (or to be run) on ARCHER/ARCHER2
- Later calls targeted at GPU codes to run on UKRI's digital research infrastructure e.g. future Exascale services, UK national AI services, national Tier-2 HPC services
 - Remit: anywhere within the UKRI's remit
 - Run via ARCHER2 contract but not restricted to codes running/to be run on ARCHER2



**Engineering and
Physical Sciences
Research Council**



**Natural
Environment
Research Council**



**UK Research
and Innovation**

What does eCSE fund?



- Not all about optimisation/performance. E.g.
 - Implementation of algorithmic improvements within an existing code in a portable manner
 - Improving the scalability of software on higher core/GPU counts in a portable manner
 - Improving a code to enhance sustainability, maintainability or usability
 - Improvements to code that allow new science to be carried out on ARCHER2/GPUs
 - Porting and optimising a code to run efficiently on ARCHER2/GPUs
 - Adding new functionalities to existing codes
 - Code development to take a code from a Tier-2 (Regional) or local university cluster to ARCHER2 or GPU system
 - Enhancing existing ARCHER2/GPU software for use in new research areas/workflows

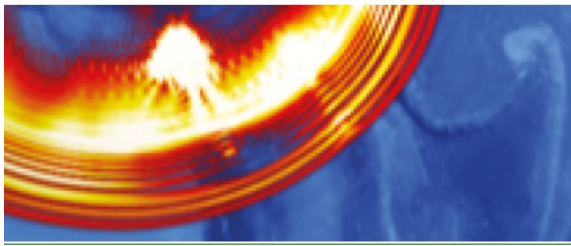
eCSE Examples



archer2

About • Support & Access • Documentation • Training • **eCSE** • Community • Research •

LOGIN TO EPCC SAFE SERVICE STATUS SEARCH



eCSE Reports

Subject Areas: Chemistry and Materials Earth Sciences and Environment Engineering and Energy Fundamental physics Mathematics and Computer Science All All by publication date

Recently published



Implementing a multiphase flow modelling capability targeting boiling in the high-fidelity software CHAPSim2
ARCHER2-ECSE08-06 Dr Wei Wang (EPIC, Daresbury Laboratory)
Subject Area: Engineering and Energy Published: 2025-05-08

This eCSE project addresses the growing need for high-fidelity simulations of boiling and related multiphase phenomena, which are essential in energy, chemical, and industrial processes. In nuclear reactors, for example, better modelling of boiling can lead to more accurate predictions of critical heat flux, improving safety margins while reducing overly conservative design constraints. This has the potential to lower construction costs and enhance power generation efficiency. Beyond the nuclear sector, understanding bubbly and boiling flows is also important for designing effective systems in chemical and process engineering. As part of this project, a new multiphase model was implemented in CHAPSim2, a direct numerical simulation tool designed for high-accuracy, high-fidelity analysis of turbulence and heat transfer in non-conventional fluids. This new model enables the software to tackle a wider range of scientific and engineering challenges. It will benefit both academic researchers and industrial engineers, providing them with an accurate, robust, and scalable framework for investigating complex multiphase phenomena. [Read more...](#)



High Performance Algorithms for the Computation of the Hardy Function - Dissemination & Development
ARCHER2-ECSE11-07 Dr David M Lewis (University of Liverpool)
Subject Area: Mathematics and Computer Science Published: 2025-02-24

The Riemann zeta-function is of fundamental interest in a number of areas of mathematical research, including number theory, theoretical physics, random matrix theory and cryptography. Hardy's z -function - defined by $Z(\sigma+it) = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{\sigma+it}}$ for $\sigma > 1/2$ - plays an important role in the theory of the Riemann zeta-function $\zeta(s)$ for $s = \sigma+it$ and its deep connection to the distribution of prime numbers. In recent years, the PI of this eCSE project discovered a theoretical formulation of the Hardy z -function in terms of generalised Gauss sums. The asymptotic characteristics of the parameters governing these sums raised the prospect of potential new algorithms for the rapid evaluation of Z , using a novel recursive methodology. The aim of this eCSE project was to bring these proposed new algorithms into reality, by developing new, faster, open access Z computational codes and make them available for general use via an open-access repository. The new source codes developed during the project show considerable speed-up compared to what was previously available, which should be of considerable interest to the mathematical research community and make an impact on practical applications of these fields to the public sphere. [Read more...](#)



GeoChemFoam - A fully-parallel pore-scale physics numerical solver package for the energy transition
ARCHER2-ECSE10-02 Dr Hannah Menke (Heriot-Watt University)
Subject Area: Engineering and Energy Published: 2025-05-01

GeoChemFoam is the world's most advanced open source pore-scale numerical simulator, and is based on the widely used Computational Fluid Dynamics code OpenFOAM. GeoChemFoam is specifically designed for research into pore-scale processes which are vital to the energy sector's transition from fossil fuels to our Net Zero future. Ambassadors of the code facilitate the decision and collaboration of carbon

- Projects produce publishable summary and more in-depth technical report
- See ARCHER2 reports:
<https://www.archer2.ac.uk/ecse/reports/>
- Earlier ARCHER reports:
<http://www.archer.ac.uk/community/eCSE/>
- Also ARCHER2 Zenodo
<https://zenodo.org/communities/archer2/>

Who does the eCSE fund?



- Funding provided for RSEs located at
 - institution of the PI
 - or RSEs from the ARCHER2 centralised CSE support team
 - or third-party institution(s)
 - ...or any combination of the above
- Funding does not cover PI or Co-I role
 - We do allow the funding RSE to take on the PI or Co-I role given enough experience/support
- Expect PI and RSEs to be UK based
- Panel assesses the team's ability to deliver on the project
 - e.g. staff skills, support from PI/Co-Is, training plan

eCSE stats



- Within ARCHER (2013-2021) and current ARCHER2 services
 - 26 calls have been run
 - 3 calls per year throughout most of ARCHER/ARCHER2
 - awarded 2105 person months
 - across 193 projects
- ARCHER2 eCSE projects
 - No. tech staff in a project: 1-3 RSEs (average 1.2 RSEs)
 - Size of project: 3-18 person months (average 10 PMs)
 - Duration: 3-24 months (average 12 months)
- GPU eCSE projects bigger (up to 36 PMs)
 - 2 calls run early 2024 and summer 2024
 - First call received 51 proposals, second call received 69 proposals (~£24M)
 - ~91 unique proposals
 - Very many good/excellent proposals
 - Able to fund just 12 projects (~£4M)
 - No further funding for any more calls right now 😞

eCSE process



- Call opens
 - Publicised via website (guidance available)/mailing lists
 - eCSE team answers questions from / meet with candidates
 - Webinar
 - Candidates submit online via website (SAFE)
 - Call closes
- Submissions assessed
 - Technically and scientifically
 - Independent panel meet to decide funding
- Awards are made
 - Successful and unsuccessful candidates informed of decisions
 - Everyone receives feedback

Early Career Panel Observers



- Alongside eCSE software development calls we have run a series of Early Career Panel Observers calls
- Early career researchers invited to on-line panel meetings as observers
 - Subject to same confidentiality/GDPR/conflict of interest rules as panel members
- Usually have 3-4 observers at each panel meeting
- So far ~50 observers have attended a panel meeting
- Good opportunity to see what reviewers are looking for
 - Intend to run alongside any future eCSE calls
- Early Career Panel Observers have gone on to submit proposals or serve on eCSE Panel

Future eCSE programmes



- Not yet known how future eCSE-type funding will be administered
- Assuming eCSE-like funding, we have control or influence over many aspects of the programme, working together with UKRI, e.g.
 - Timelines/frequencies of calls
 - Process (e.g. from working with applicants wanting to apply through to assessment and award of funding)
 - Make up of panel
 - Guidance for applicants
 - Guidance for panel (including assessment criteria)
 - Size of awards

DiRAC RSE



- DiRAC provides HPC resources to UK's Scientific Research Communities in theoretical cosmology, nuclear physics, astrophysics, particle physics, and solar and planetary science.
- Possible to apply for RSE effort within yearly calls
- Effort comes from existing RSEs within DiRAC
- 5 projects funded from April 2026 (~39 PMs) – plus on-going projects from previous years

UK's AI Factory Antenna at EPCC (UKAIFA)



- Aims to boost AI adoption among businesses and researchers
- UKAIFA is collaborating with the HammerHAI AI Factory, coordinated by HLRS
- Wide range of targeted end users: start-ups, small/Medium-sized enterprises (SMEs), larger industrial companies, public sector
- Focus on a region-by-region approach in UK as project progresses
- Direct engagement activities e.g. roadshows, workshops, etc
- There will be a regular calls for applications
 - Effort provided by RSEs at EPCC

Interactive Session



| epcc |

Aims of session

- ~30 mins: short talks of the eCSE and RSMF programmes
- ~45 mins of interactive session – we will give you some questions to consider (not restricted to these!)
- Overall aim is to collect feedback on the points raised in the questions
- Additionally we will gather list of priorities based on discussion
- Final aim (depending on time) – Q&A session at end where speakers can comment on possible implications
 - Based on knowledge of recent, current landscape, etc.
- Funding comes from RCs and exactly what to fund and when is at the UKRI level – however, we do have a voice and some say in how calls are run

Questions



- Is the remit of the calls suitable?
 - E.g. hardware focus, type of work funded, etc.
 - Related question: Are there particular areas or particularly communities the calls should be focussing on?
- Is the range of suitable award sizes suitable for what the community needs.?
 - E.g. length of funding, amount of RSE effort
 - Related questions: Is the frequency of calls appropriate? What should the funding include (Project Lead overheads? Travel?)
- Is the submission and reviewing process suitable for such calls
 - peer review, single blind, no. reviewers per proposal, technical input, feedback, etc.
- How should environmental sustainability be integrated into calls?
 - How should this be done?
- Anyone who is online only (i.e. not in the room) and wants to join the discussion – please note this in the chat so we can involve you!
- Please discuss the above points – try to come up with what you consider to be the most important points

Interactive Session

Feeding back from the breakout groups



Feedback from session



- What are the priorities for future soft dev funding in terms of
 - Remit of call
 - Size of awards
 - Process
 - Environmental stability

Interactive Session

Thank you for taking part

